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Developing new product service systems (PSS): methodologies and operational tools

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Abstract

The co-evolution of industrial production and social patterns calls for systemic solutions that can only be provided by partnerships between companies and other stakeholders, including final users. Such partnerships are defined as Solution Oriented Partnerships (or SOP). Product Service Systems (PSS) are the catalyser of such solutions. The capability of PSS to become an attractive solution depends on factors that are commonly considered to belong to the design domain. The role of designers is therefore essential to the definition of effective and attractive PSS. Designers are now urged to find their own methodological approach to the design of PSS. This paper addresses this need by proposing methods to define a map of the actors involved in PSS, methods to define requirements and structure of a PSS and methods to represent and blueprint a PSS.
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Keywords: Solution oriented partnership; Product service systems design methodology; Representation of Product service systems

1. Introduction

Industrial production is evolving towards models that more adequately address an epochal shift from mass consumption to individual behaviours and highly personalised needs. Such an evolution is often facilitated by rethinking the industrial offering, from the production of goods to the provision of systemic solutions consisting of products and services.

This shift has widely been recognised in the management and marketing disciplines, and is now becoming part of the wider horizon of the design discipline.

The role of designers in this shift is indeed very relevant, as many systemic solutions are only possible when different actors (companies, institutions and final users) join their effort to solve common problems and achieve common goals. Within the Suspronet network, the partnerships created by the convergence of different stakeholders for the generation of the solution have been defined as Solution Oriented Partnership.[1] The glue of such partnership is attractive design solutions based on a mix of material and immaterial components, which satisfy the requirements of each of the stakeholder. Such solutions are commonly described as Product Service Systems (PSS).

The role of designers in such partnership is therefore critical, although the definition of specific methodologies to manage some critical aspects of the design process of PSS has rarely been considered in design-related disciplines.[2]

[1] The concept of SOP has been widely discussed in several conferences organised by Suspronet, a network of industries and institutions, focusing on the design of products and service for a sustainable and competitor growth. For a clearer understanding of characteristics and potential of SOP see the contribution of Jegou et al. to the most recent Suspronet conference [1].

[2] It is worth stressing the difference between a *methodology* and a *method*. A methodology defines *an operative paradigm*, i.e. a "toolbox" including several different methods and tools that can be used to solve determined logical or operational problems [2]. PSSs represent a very wide area of intervention for a designer. The definition of a standard set of methods and tools to use to design PSSs is therefore impossible. However, designers should consider creating their own toolbox including methods and tools to be used in different contexts and for different PSS.

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Several contributions from marketing and management disciplines [3–8] are a good starting point for such a definition; however, the innovation potential of PSS requires an accurate design process, which consider product design issues, as well as communicational, social and economic aspects.

A PSS is a social construction, based on ''attraction forces'' (such as goals, expected results and problem-solving criteria) which catalyse the participation of several partners. A PSS is the result of a value co-production[3] process within such a partnership. Its effectiveness is based on a shared vision of possible and desirable scenarios.

The design activity within this process should therefore focus on the catalysing factors that generate cohesion, which means that designers should have tools to:

- Work on the identification of the actors in the network, on the basis of defined analytical frameworks.
- Work on possible PSS scenarios, verifying use cases, sequences of actions and actors' role; defining the requirements for a PSS and the logical and organisational structure of PSS.
- Work on possible representation and management tools to represent a PSS in all its components, i.e. physical elements, logical links and temporal sequences.

The following sections will refer to several research experiences in cooperative research works and in academic curricula, developed by the author in cooperation with different research and teaching teams.

Tools and methods discussed in this paper are not new; they are commonly used in other disciplines. The contribution of this paper, however, consists in the application of such tools in design exercises developed within cooperative research projects and in academic curricula,[4] with the aim of verifying them and generating a methodology for the design process of a PSS.

2. Identifying the actors network

Industrial products and services are not only a technical entity, but also the result of a socio-technical process. Innovation process and the trajectory of such a process are strongly influenced by the actors who directly or indirectly participate to them [11,12]. This means that the design activity should be based on the convergence between several social and technological factors, including:

- the technological knowledge embedded in the artefacts used for the service; and
- the social, technological and cultural frames of the actors participating in or influencing the development of the system.

The combination of such a heterogeneous mix of elements (people + cultural frames + technological artefacts) suggests that the designer has the function of linking technological artefacts to the attitudes of relevant social groups in order to accept or reject certain products and technologies.

Technological artefacts and infrastructures often reveal the strong influence of the socio-technical culture of their designers/developers. Their cultural frames are intelligible through the physical and technological characteristics of the artefacts. This is very relevant to the development of a PSS: severe limitations to the development of certain characteristics of PSS emerge when such characteristics are beyond the socio-technical horizon of the developers of the technological infrastructure. Such limitations are even more evident when the PSS is based on high levels of automation.

> Within the Telecentra project,[5] for instance, the design team encountered severe limitations in the use of remote file exchange and sharing, due to the internet settings in each physical location and to several firewall systems. The communication system was mainly designed having in mind several (justified) security issues. At the time of the development of the project, though, the needs of nomadic workers to access and communicate through the Internet often transcended such security requirements. This eventuated in a critical limitation in the design of PSS for this category of workers.

Relevant social groups, according to Bijker et al. [11] are not only those groups that actively participate to the development of the product-service system, but also those groups and actors that indirectly participate in such a process or even those actors that may oppose to the PSS. Such a perspective helps defining a complex picture of the scenario in which the PSS is supposed to be developed.

> Within the TeleCentra project, for example, the development of a neighbourhood centre in a school was strongly influenced by the needs of school personnel that was not participating in the project and sometimes was not even aware of the presence of the centre in the school.

In order to facilitate the analysis of relevant social groups, Bijker [12] provides a set of reference parameters each group refers to when shaping innovative solutions (Table 1). Bijker's

[3] The concept of value co-production in service activity has been introduced by [9,10].

[4] This paper mainly refers to the TeleCentra project, coordinated by the author within the framework of a cooperative research project between RMIT University and some Australian companies and to some projects developed as part of the teaching activities on system design at the School of Architecture and Design at Aalborg University.

[5] The TeleCentra project consisted in the design of a PSS for nomadic office workers, i.e. for people who spend the major part of their working time far from their office. The project was articulated in three subprojects: the development of a neighbourhood centre, the development of a physical office and the development of a virtual office. While the neighbourhood centre was developed for a school, the physical office and the virtual office were developed to the pre-commercial phase with two different companies. The neighbourhood centre and the physical office had a test run (for the physical office users were also required to pay a small fee for accessing the service).

Table 1
Framework of criteria for the analysis and individuation of relevant social groups [5]

Goals	The needs each group wants to satisfy in relation to specific activities
Key problems	The problems perceived to be relevant in relation to specific activities
Problem-solving strategies	The strategies considered admissible and effective in solving the main problems
Requirements to be met by problem-solving strategies	Admissibility and effectiveness criteria for problem-solving strategies
Current theories	Theoretical knowledge supporting the activity of each group in setting goals, identifying and selecting problems and proposing admissible problem-solving strategies
Tacit knowledge	Practice based knowledge upon which each group relies to set goals, identify and select problems and propose admissible problem-solving strategies
Testing procedures	Procedures used to evaluate the effectiveness of each problem-solving strategy
Design methods and criteria	Methods and parameters used for proposing technological solutions to emerging needs
Users' practice	Users' attitudes towards existing solutions to the present needs
Perceived substitution function	Products, services or sets of functionalities each group believes to be replaced by the proposed PSS
Exemplary artefacts	Products and services that are used as models in developing new solutions. Often deriving from the perceived substitution function

parameters can be used to generate different profiles of the possible users of a service.

On the basis of such profiles it is possible to generate interaction maps between the actors in the system, such maps may focus on the layers of interaction (Fig. 1) or on interaction scenarios (as in Figs. 2–4).

The map in Fig. 1 emphasises direct and indirect relationships between the actors in the system, it also points out the dependence of the system from infrastructural conditions, some of which concern high decisional levels and cannot be changed in any way. The design process in this case should consider such conditions as external to the system and focus on design solutions that fit in such external conditions.

The maps in Figs. 2–4 focus instead on the interaction between social groups, emphasising each group's involvement in the system. This kind of representation can also be used in design phases, in order to figure out the interaction between different social groups according to different organisational and social scenarios.

Further specification of interaction maps in the design phase of PSS may lead to a broad definition of the PSS blueprint as in the work reported by Jegou et al. on the HiCS project [1].

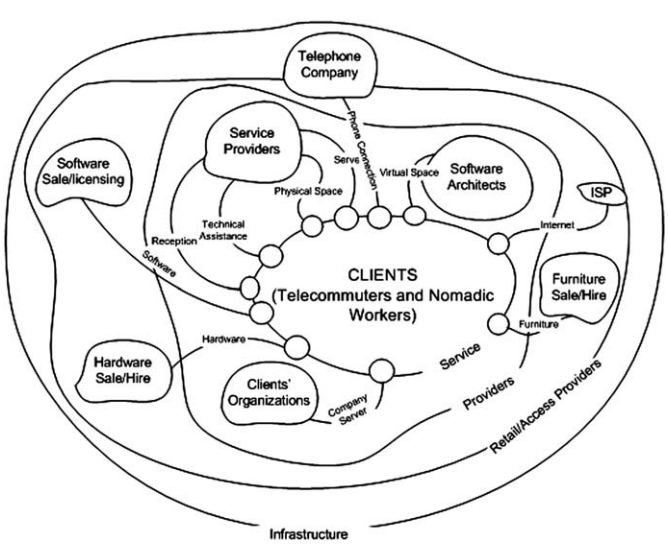

Fig. 1. Map of interaction in a PSS. Source: TeleCentra project.

3. Envisioning the PSS: scenarios and use cases

The existing tools and methods available to the designers are very adequate to control a product-focused design process, where the major part of the factors concurring to the process can be objectively defined. The design of a PSS, instead, falls in a different domain. The design process in this case focuses on systemic aspects and is based on the assumption that its final result is co-produced by a network of social actors.

The design discipline has no methodologies to operate in such domains. While the development of the physical features of a product is based on an exploration of dimensional, aesthetical technological and mechanical characteristics of the product, the service components in PSS introduce new variables. The new variable includes the time dimension, the dimension of the interaction between people, and other hidden dimensions related to cultural mind frames and social habits.

Some tools and methods are available in other disciplines, which would help managing the complexity of the design process.

A helpful tool that may support a systemic approach to the design of PSS is IDEF0 (Integration definition for function modelling). This tool, mostly used by system engineers, may help covering areas of the design process characterised by a complex systemic structure. IDEF0 is a modelling technique that allows for progressive detailing of the functions and actions in the system, while keeping the link between each element in the system. The system is modelled as a series of boxes, representing a function of the system. Arrows entering the left side of the box are inputs. Inputs are transformed or consumed by the function to produce outputs. Arrows entering the box on the top are "controls" that specify the conditions required for the function to produce correct outputs. Arrows leaving a box on the right side are outputs, i.e. the data or objects produced by the function. Arrows connected to the bottom side of the box represent mechanisms, i.e. means

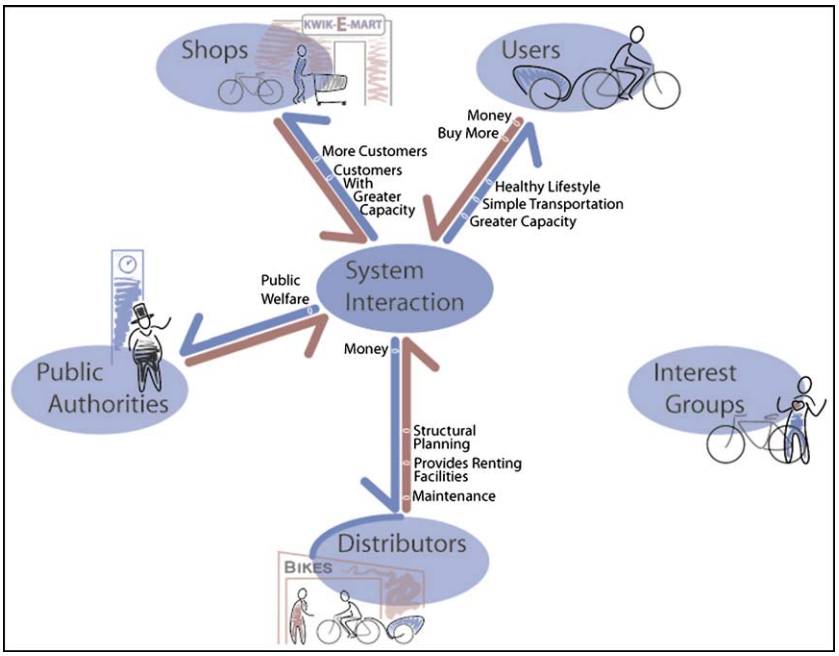

Fig. 2. Interaction map for a shared bicycle trailer system, System concept 1. The service is offered by bicycle shops. The map describes the nature of participation of all the actors in the system. Source: "My Way", project for the 7th-semester ID at Aalborg University.

that support the execution of the function or links between models or portions of the same model.

Each of those boxes can be decomposed in a hierarchy of sub-boxes, which can be analysed with the same logic (Fig. 5).

This tool provides an accurate representation of logical, time related and physical connections between various phases and components of the system, with the possibility of recomposing the systemic view in any moment, by referring to the boxes in the highest order of the hierarchy.

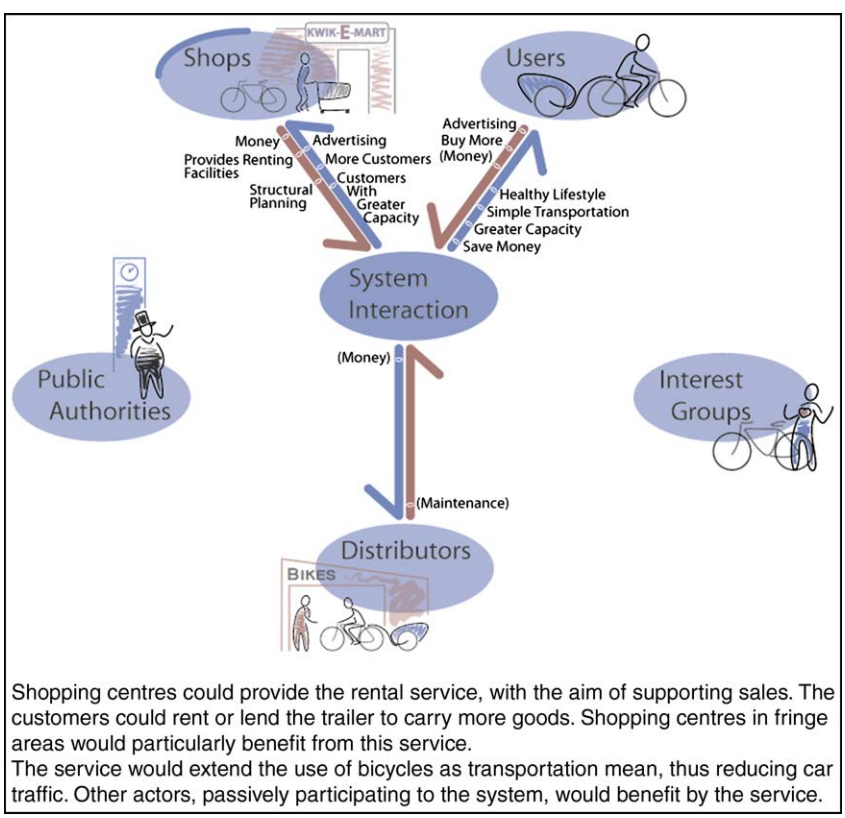

Fig. 3. "My Way", System concept 2. The service is provided by shopping centres.

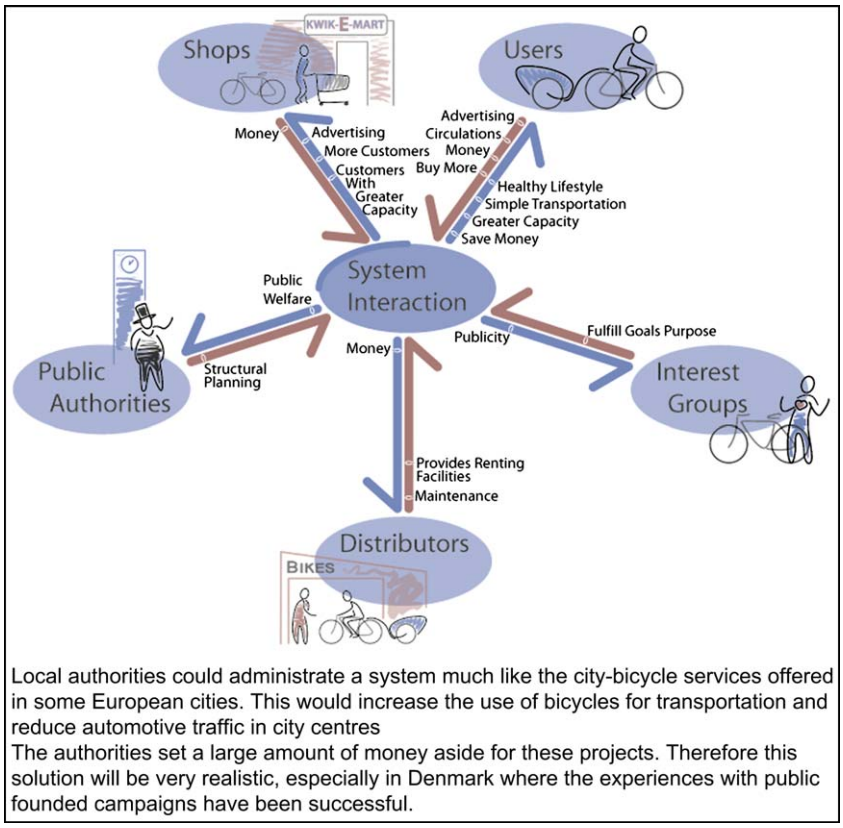

Fig. 4. "My Way", System concept 3. The service is promoted by the local government.

IDEF0 is obviously covering a part of the methodological problems related to the systemic nature of PSS. This tool is very adequate in systems with a higher grade of predictability. A higher grade of subjectivity (e.g. variation of actor's behaviours) and unpredictability (e.g. variation of contextual conditions) may bring about an infinite number of configurations and situations this technique cannot cover.

IDEF0 has been used and tested in several projects by industrial design students at Aalborg University. The tool proved to be very helpful in representing production systems with limited ranges of users' choices (e.g. a function in a hospital, an information service for train travellers), but too complex to manage, when dealing with PSS characterised by a wide range of possible users choices. (e.g. a toy library, a car-sharing system).

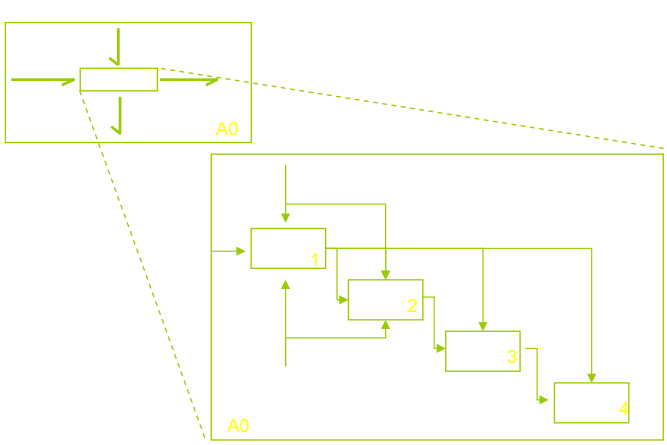

Fig. 5. A schematic representation of the modelling logic used in IDEF0.

A more complete methodological picture can be outlined when IDEF0 is used together with other techniques such as scenarios and use cases. They are used in information technology to develop the architecture of information systems [13,14]. The use of such techniques in design discipline would help eliciting requirements for the PSS, but also would provide a broad picture of the PSS configuration, which can be eventually defined with other systemic techniques.

The development of scenarios is based on the actors' profiles and on the possibility that different actors are involved in different configurations of the PSS. Each scenario is composed by a number of descriptions of events (use cases) that describe the details of sequence of action for each function included in a scenario.

Once defined through scenarios and use cases, a PSS needs to be thoroughly explored, in order to understand all the phases in which the designer's intervention is needed.

The requirements emerging from use cases can be accurately analysed and decomposed, in order to work out the nature of the design task, by doing this it is possible to understand which product and service component are need for each requirement (Fig. 6).

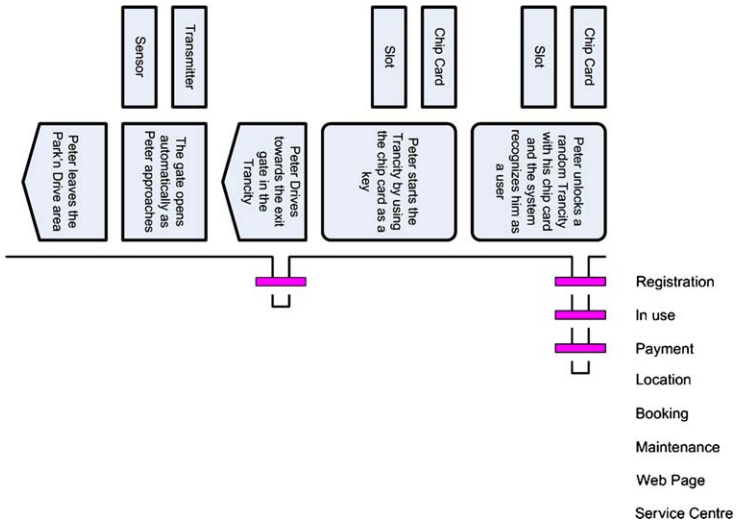

Fig. 6. "Trancity": graphic representation of requirements for a use case of a car-sharing system.

4. Representing the structure of PSS towards a blueprinting technique

The graphical representation of a product is an important component of the design activity and is critical in a context of industrial production; a product blueprint makes it possible to reproduce the product according to the designer's intention. The design discipline, however, has never developed graphical tools for PSS to a standard comparable to the blueprinting techniques in product design.

The relevance of a graphic representation of a PSS is even clearer when considering the catalysing role of PSS in SOP. In such a context the communication of characteristics of the structure of PSS is fundamental for a common understanding of the nature of the design solution.

The graphical representation of PSS is the topic of an ongoing debate, with several interesting contributions, but no final solution about the definition of a standard for blueprinting PSS.

A solid ground for the development of a blueprint of PSS can be represented by the techniques proposed in the previous sections, some of which (e.g. IDEF0) are intrinsically based on graphic notations.

Graphic notations are also used in information technology to illustrate use cases. The diagrammatic representation used in information science, however is not sufficient to represent all the elements involved in the design of a PSS. While information science focuses on logical and relational factors, the design discipline also considers any element that influences the perceived quality of a PSS, including space, time, movement and physical layout. A more complete graphical description may take into account Shostack's [7] suggestions for service blueprinting, which includes indications of what happen beyond the *line of visibility* that separates front office from back office. Further developments of this technical representation have been proposed by the author of this paper, as shown in Fig. 7 [15,16]. The figure includes information about physical and virtual spaces, front and back office, characteristics of the agents (i.e. whether an action is performed by a person or a machine), space and movement.

Graphical representations based on use cases provide a detailed description of the PSS to be designed and may be compared with technical detailing in product design. A more comprehensive representation of the PSS, including the definition of the main components and the interactions among them can be provided by project plans, as developed in the HiCS project.[6]

5. Conclusions

Although PSS are fundamental for the development of solution oriented partnerships, and consequently for sustainable solutions, the design discipline has not yet defined an operational paradigm, i.e. a set of standard tools and methods, to design and develop PSS. This paper contributes to close this gap by proposing methods and tools that have been effectively used in some previous project. However the methodology emerging from this paper is far from being complete and commonly accepted. This is only the very first step to support a PSS design process within the context of SOP.

The application of those tools may be different from case to case. The intrinsic complexity of some PSS requires that such tools to be used with a high degree of flexibility: "narrative" tools, such as scenarios and use cases should be preferred in the definition phases, whereas more "technical" tools are preferable for defining the structure of PSS. Furthermore different working groups may prefer narrative tools or more technical

[6] The methodology developed within the HiCS project has been illustrated by Francois Jegou, Luisa Collina and Ursula Tirshner in different contributions for the Suspronet network. The methodology is being published as part of the final outcome of the project.

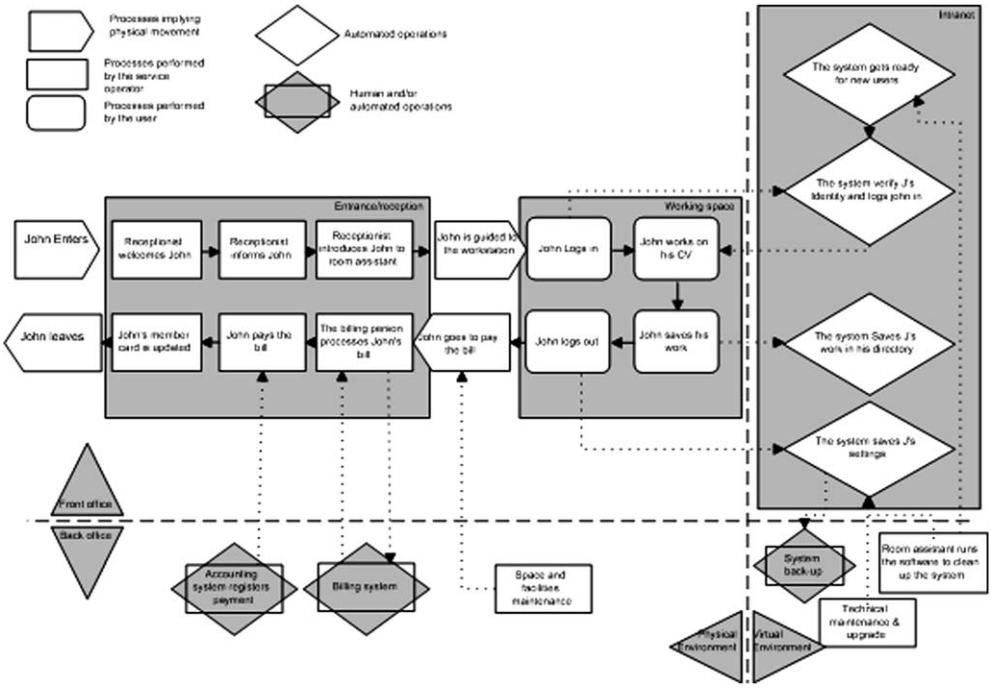

Fig. 7. Service blueprint including indications on actors, movements and environments. Source: TeleCentra project.

methods according to their approach and the characteristics of the participants to a SOP.

It is worth emphasising, however, that the discussion about a methodology to design a PSS is still open and is critical for the development of sustainable solutions. A comprehensive and unique methodological approach is probably impossible in this area, where the margin of uncertainty about contextual conditions may be very high. New case studies and further applications and improvements of the proposed methods, however may contribute to define a clearer methodological approach to the design of PSS.

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